

Mechanical and Thermal Behavior of Infill Pattern Variations in Fused Deposition Modeling (FDM) based 3D Printed Parts

Andrew Walsh



Sukhpreet Singh



Dilli Babu
Kalluru

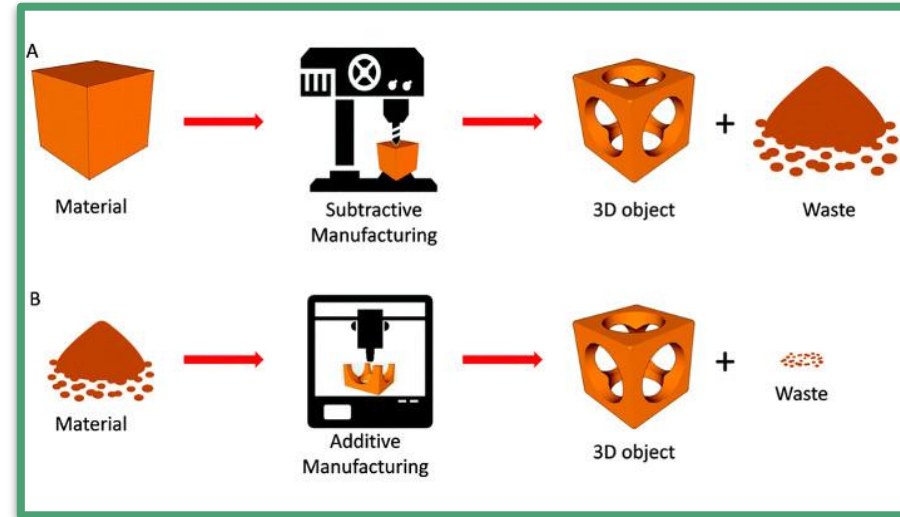


Vishavjit Singh Khinda



Additive Manufacturing

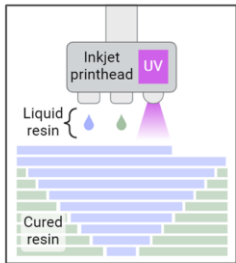
- process of creating 3D objects by adding material layer by layer based on digital designs, in contrast to traditional manufacturing methods which often involve subtracting material (like milling and drilling)
- Applications
 - Prototyping (rapid and low-cost iteration)
 - Aerospace, automotive, medical (custom parts manufacturing)
 - Education and research



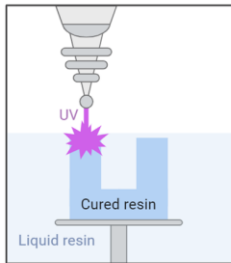
Types of Additive Manufacturing

- **Material Jetting** – Tiny droplets of material are deposited layer by layer, similar to inkjet printing.
- **Stereolithography (SLA)** – Uses a laser to harden liquid resin into solid layers.
- **Laser Powder Bed Fusion (LPBF)** – A laser or electron beam fuses powdered material into solid parts
- **Binder Jetting** – A binding agent (like glue) is selectively deposited to join powder particles.
- **Directed Energy Deposition (DED)** – Material is melted as it's deposited, often using a laser or electron beam.
- **Material Extrusion (e.g., FDM)** – Filament is heated and extruded through a nozzle to build parts
- **Sheet Lamination** – Sheets of material are cut and bonded together to form an object.

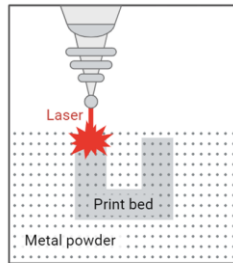
Material jetting



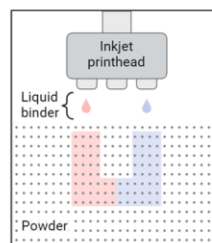
Stereolithography



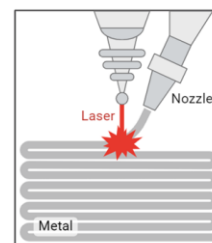
Powder bed fusion



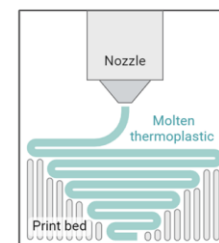
Binder jetting



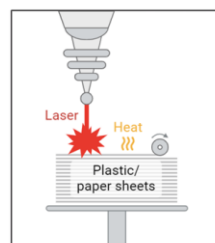
Directed energy deposition



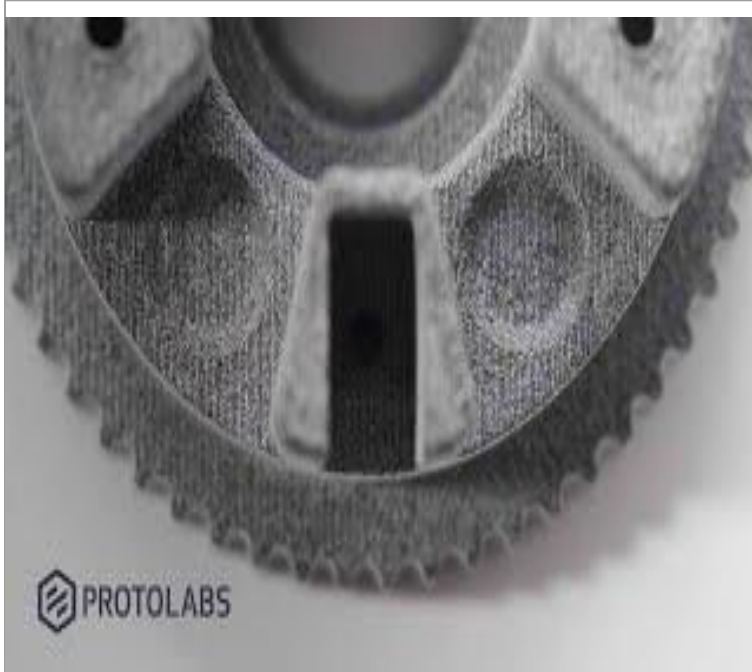
Material extrusion



Sheet lamination

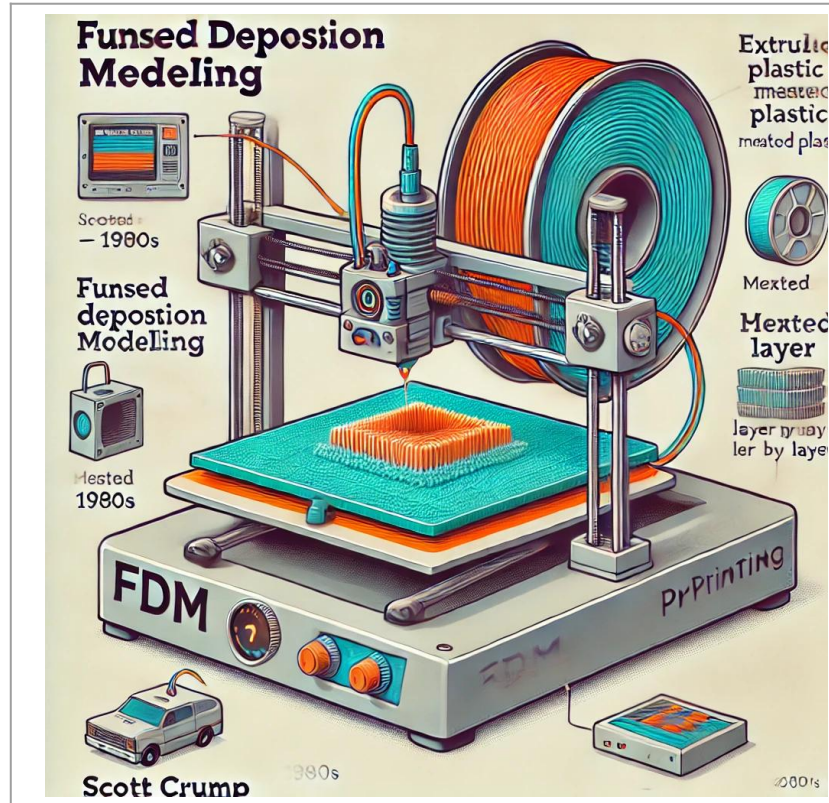


FDM 3D Printing



In the world of additive manufacturing, Fused Deposition Modeling (FDM) stands out as one of the most popular and widely used 3D printing technologies. From prototyping to final product creation, FDM has transformed the way we design and produce objects.

What is Fused Deposition Modelling (FDM)



Fused Deposition Modeling, which is more popularly known as FDM, is a 3d Printing technology that builds an object layer by layer with thermoplastic materials. This was developed in the 1980s by Scott Crump. In the process of FDM, a plastic filament is heated up to its melting point and comes out of a nozzle as droplets that form layers upon each other to produce three-dimensional objects. FDM is known to be the most accessible to use. This makes the process ideal for both the professional and hobby worlds.

Fused Deposition Modelling



- FDM is 3D printing technology which is mostly used for modelling, prototyping, and production applications.
- This makes it perfect for demanding designers and engineers in aerospace, medical, electrical, consumer goods, and other industries.

Fused Deposition Modeling (FDM) process used in 3D printing

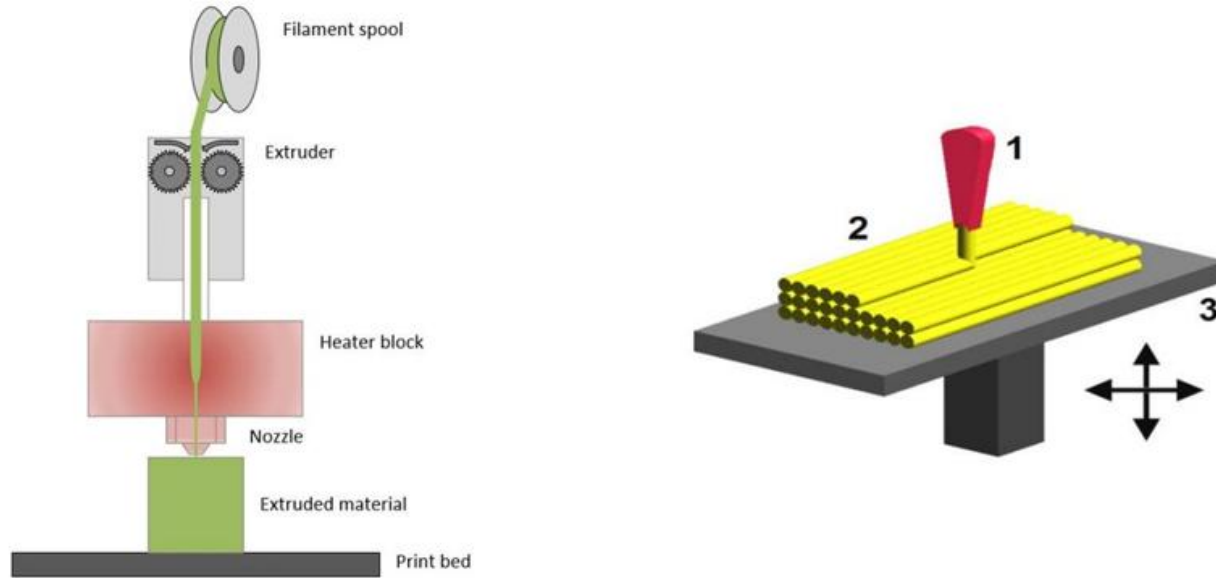
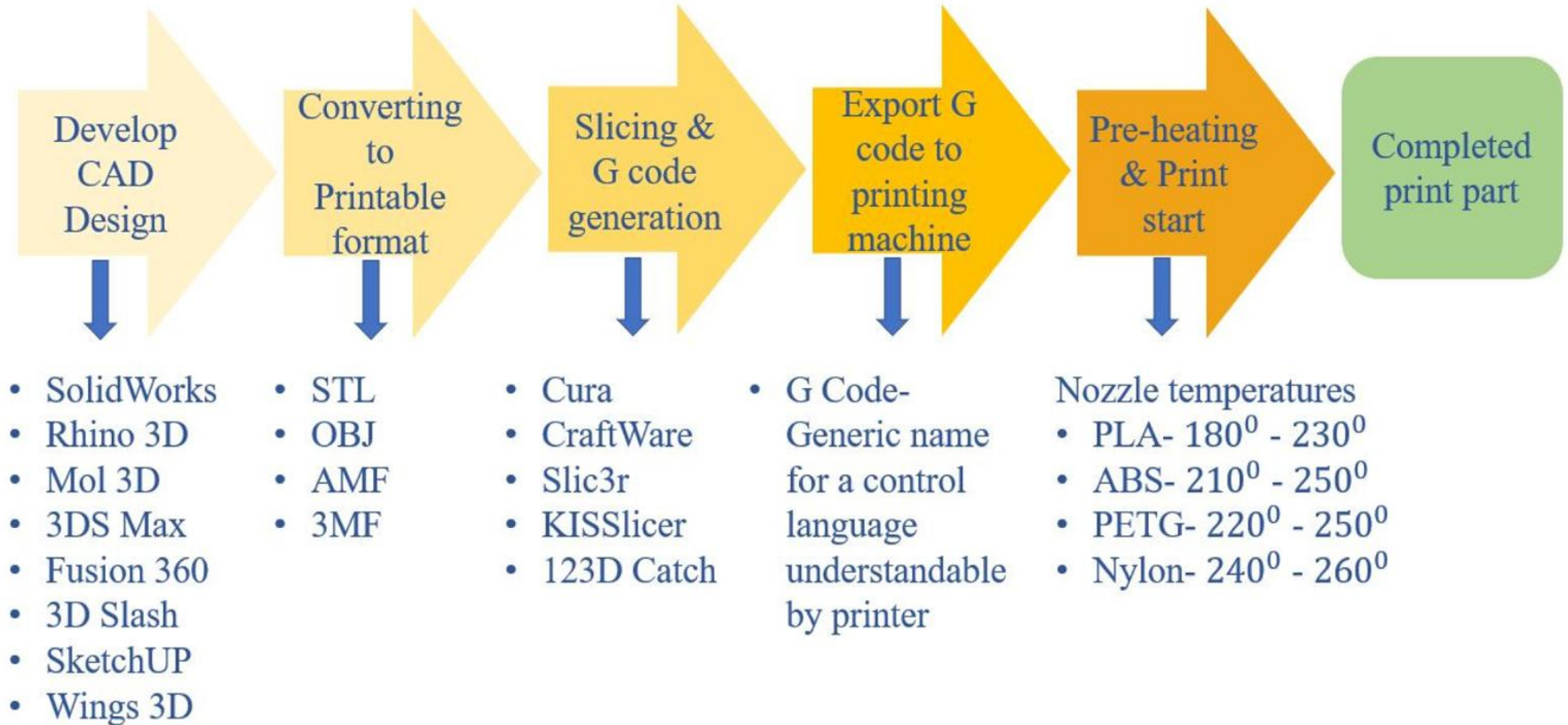
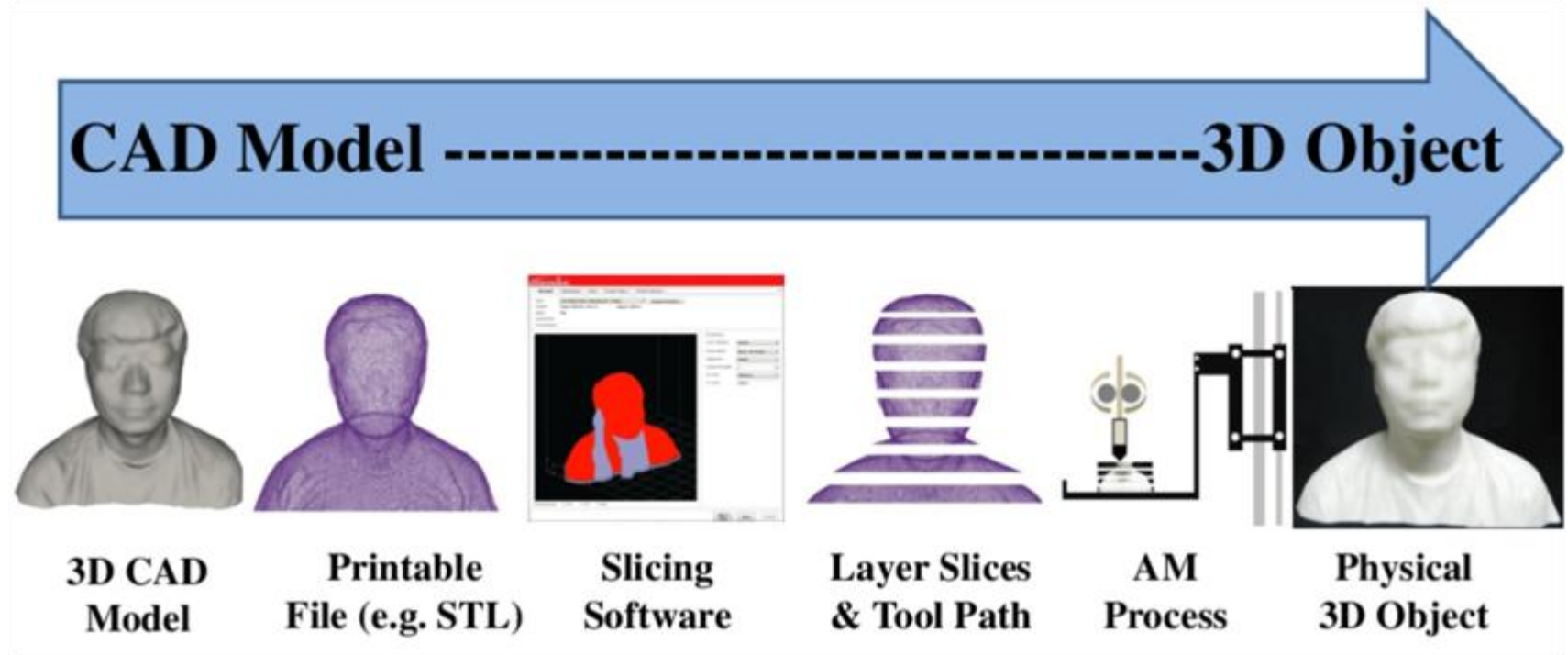


Fig. 1. Schematic model of FDM process (1-nozzle, 2-building material, 3-print bed)

3D Printing Process Overview



3D Printing Process Overview



Advantages of FDM

1. Cost-Effective

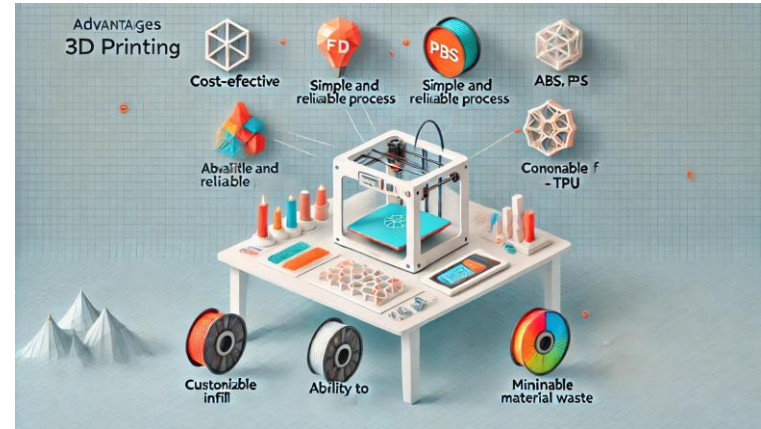
FDM printers and materials (thermoplastic filaments like PLA) are relatively inexpensive, making the technology accessible for hobbyists, educators, and industries.

2. Simple and Reliable Process

The FDM process uses a straightforward mechanism-melting and extruding filament through a nozzle -which ensures ease of use and maintenance.

3. Wide Range of Materials

FDM supports various thermoplastics such as PLA, ABS, PETG, TPU, and composites, allowing for versatile applications.



4.Ability to Produce Complex Geometries

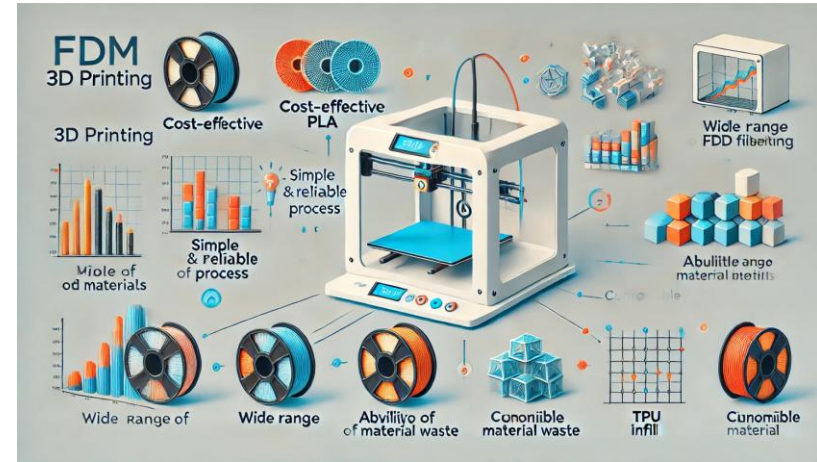
By depositing material layer-by-layer , FDM can create intricate shapes and internal structures that are difficult or impossible with traditional manufacturing.

5.Customizable Infill Patterns and Densities

Users can adjust infill patterns and densities to optimize strength, weight, and material usage, which is especially useful for functional prototyping.

6.Minimal Material Waste

Since FDM is an additive process, it only uses the material necessary to build the part, reducing waste compared to subtractive methods.



FDM Printing Parameters

Most commonly adjusted settings

- **Layer Height**
 - Thicker Layers: Stronger parts, faster printing
 - Thinner Layers: Higher accuracy
 - Achievable range is dependent on nozzle size
- **Infill Density**
 - Higher Density: Stronger parts
 - Lower Density: Faster printing, less material
 - Typically no more than 30-50%
- **Temperature**
 - Nozzle Temp: Effects max speed, layer adhesion, surface finish, overhang capabilities
 - Print Bed Temp: Effects part adhesion and warping
 - Temp range is primarily dependent on material
- **Infill Type**
 - Lots of different infills that can affect print time, strength, and stiffness

	Time	Percentage	Used filament	
Perimeter	1h16m	19.9%	235.09 in	17.81 g
External perimeter	2h7m	33.2%	240.53 in	18.22 g
Overhang perimeter	2m	0.4%	3.05 in	0.23 g
Internal infill	1h21m	21.2%	194.01 in	14.70 g
Solid infill	35m	9.3%	114.18 in	8.65 g
Top solid infill	5m	1.4%	10.47 in	0.79 g
Bridge infill	11m	3.0%	23.75 in	1.80 g
Skirt/Brim	6m	1.6%	9.21 in	0.70 g
Custom	6s	0.0%	0.85 in	0.06 g
Estimated printing times [Normal mode]:				
First layer: 11m				
Total: 6h21m				
Show stealth mode				

Common FDM Materials

PLA

- Easy to print
- Very inexpensive
- High dimensional accuracy
- Moderate stiffness
- Moderate strength
- Most common material in hobby 3D printing and scale models

ASA

- Difficult to print
- Moderately priced
- UV and temp resistant
- Moderate to high stiffness
- High strength
- Most common in outdoor and high heat applications

PETG

- Easy to print
- Moderately priced
- Durable
- Water and chemical resistant
- Semi-stiff
- Moderate strength
- Most common in water resistant and snap fit components

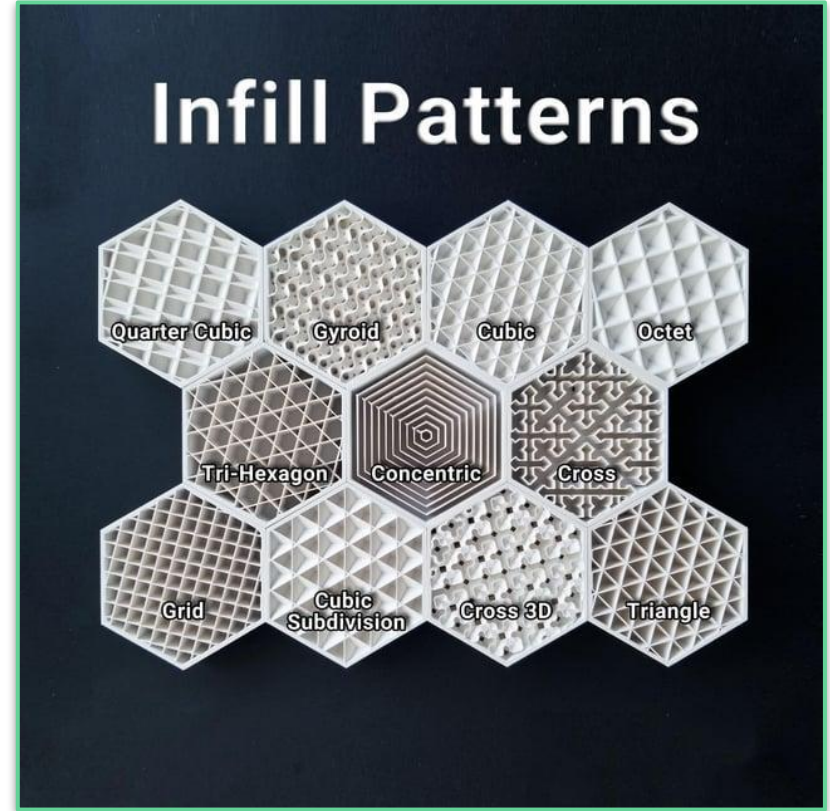
Other FDM Materials

- **ABS** - Very durable. Great for strength and high temp applications
- **TPE/TPU** - Very flexible. Good for gaskets, tires, vibration dampers
- **HIPS** - Dissolvable in water. Good for difficult to remove supports
- **Carbon Fiber** - Added to other materials to increase strength
- **Polypropylene** - Semi flexible. Great for high cycle parts



Infill Patterns

- Many different patterns offered by slicers/companies
- Primarily affect strength, stiffness and print time
- No general consensus to the best infill pattern
 - Gyroid is very common and can be used in most applications
 - Things to consider: direction of force, stiffness, speed

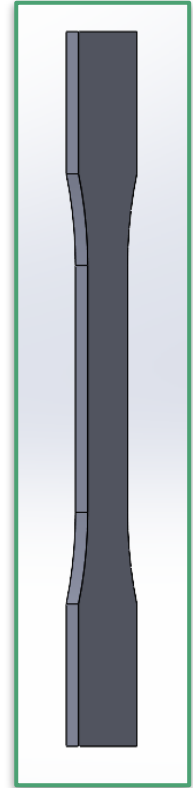


Project Setup

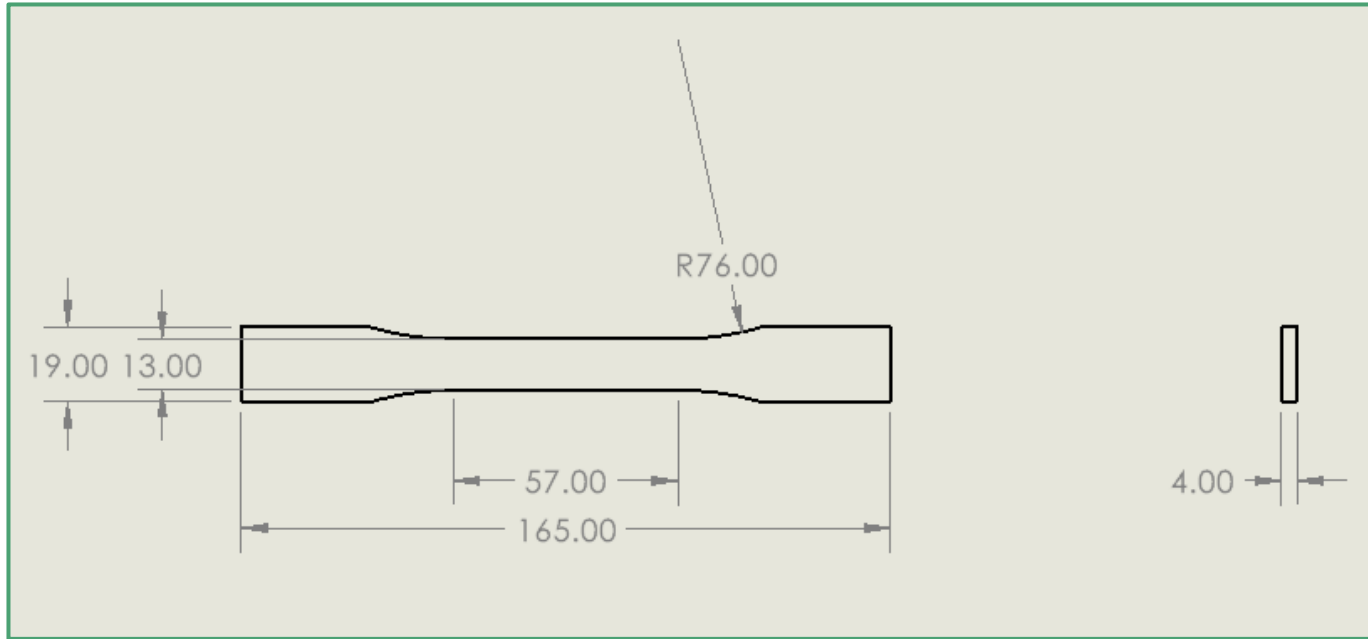
Tensile Test Part

Object was modeled in Solidworks and imported into Ansys

- Tensile Simulation
 - ASTM D638 Type I
 - 0.2 mm layer height
 - 0.42 mm layer width
 - 30% infill
 - 2 top and bottom layers (.4 mm)
 - 2 walls (0.84 mm)



Tensile Test Part - ASTM D638 Type I

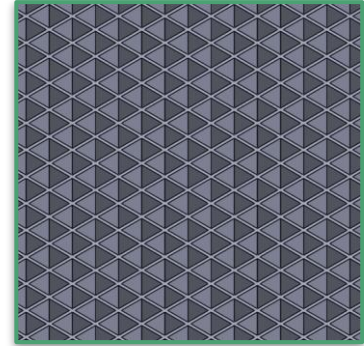
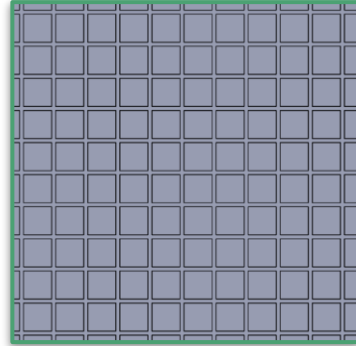


Part Infills

Infill patterns were manually modeled in Solidworks

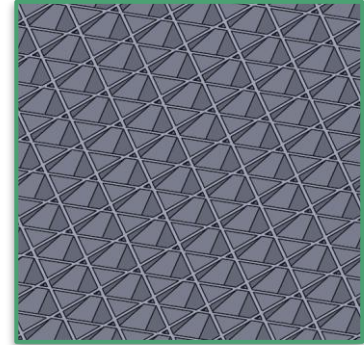
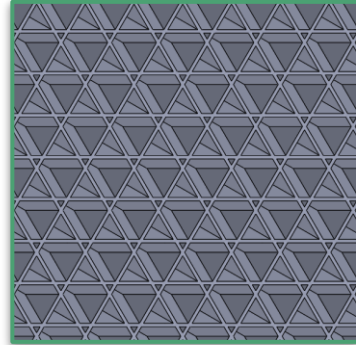
Infill 1 - Rectilinear

- 2D Infill Pattern



Infill 2 - Cubic

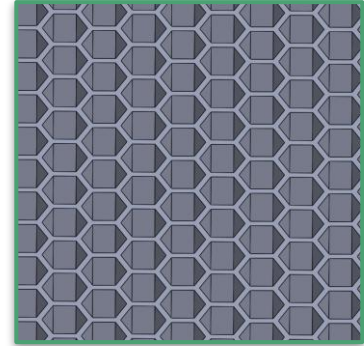
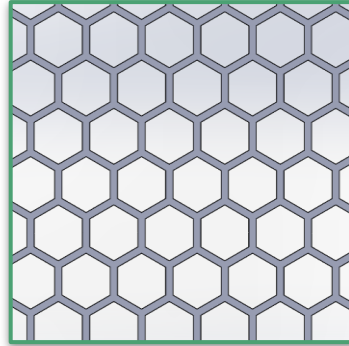
- 3D Infill Pattern



Part Infills - Cont

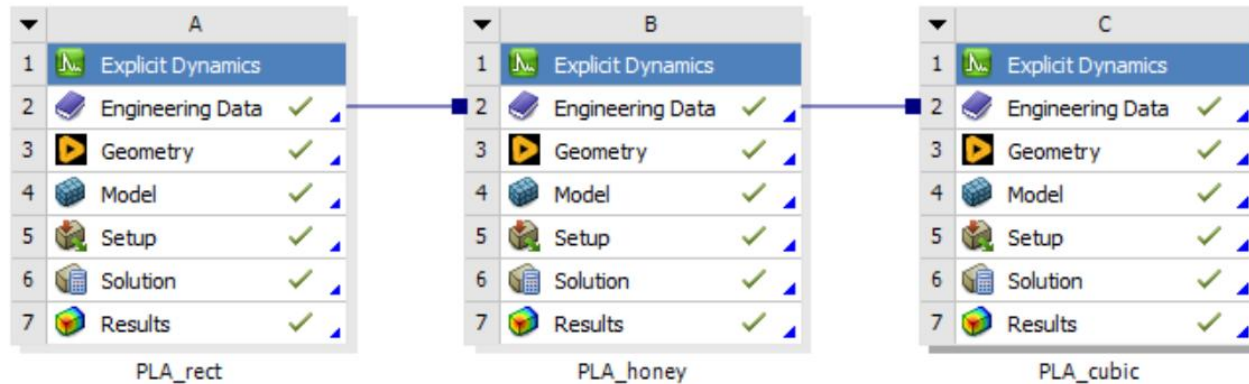
Infill 3 - Honeycomb

- 2D Infill Pattern



Structural Analysis: Ansys Study Setup

- Explicit Dynamics Study was utilized to conduct tensile tests for 3 different infill patterns
- First, the Engineering Data was generated for PLA using datasheets from manufacturers and experimental data from previous research papers



Structural Analysis: Ansys Study Setup

- Stress Strain curve was plotted using Multilinear isotropic hardening

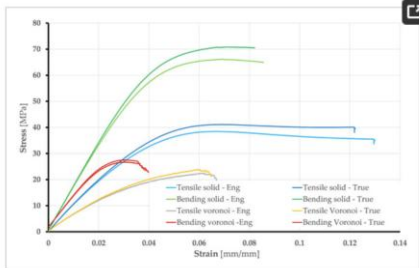
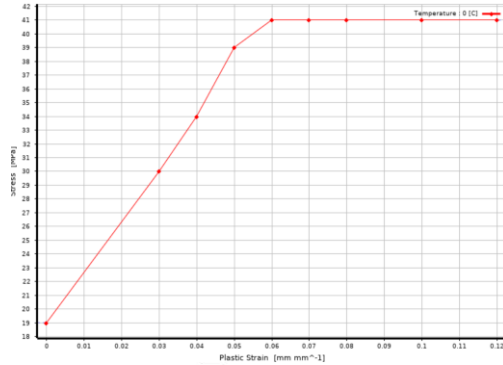
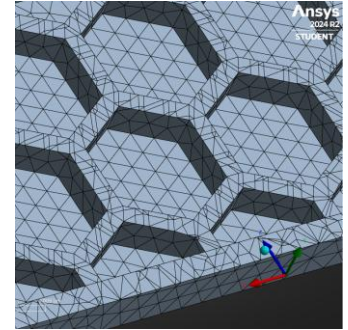
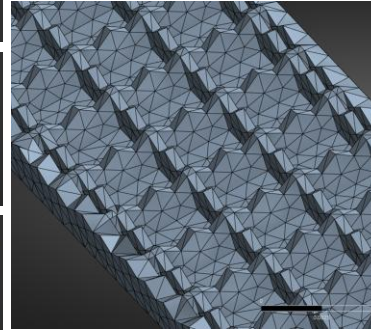
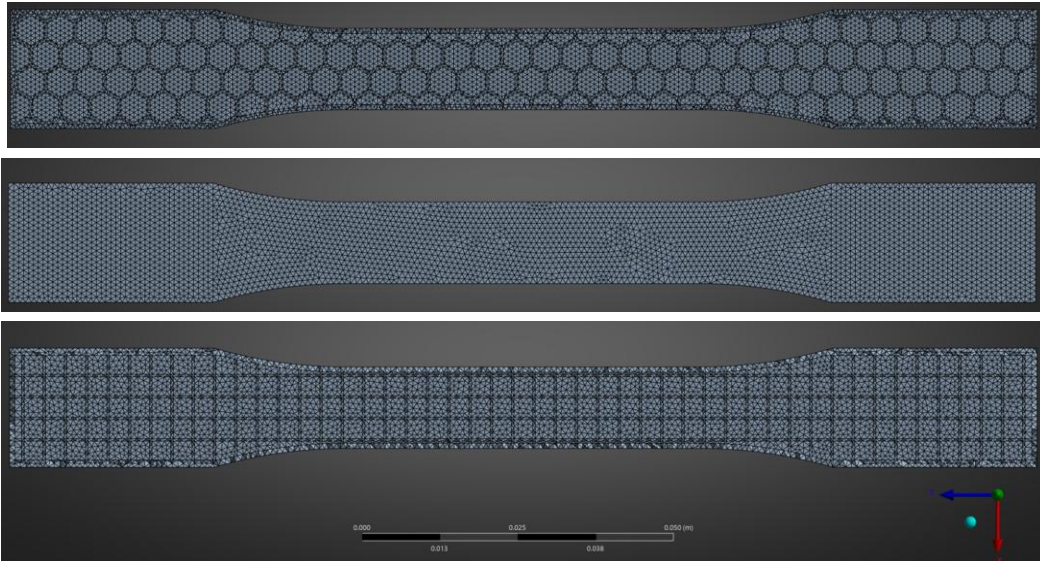


Figure 8. Stress vs. strain average curves: engineering and true.

Properties of Outline Row 3: PLA			
		B	C
1		Value	Unit
2	Material Field Variables	Table	
3	Density	1240	kg m ⁻³
4	Isotropic Elasticity		
5	Derive from	Young's Modulus and Poisson's Ratio	
6	Young's Modulus	3.5	GPa
7	Poisson's Ratio	0.35	
8	Bulk Modulus	3.8889E+09	Pa
9	Shear Modulus	1.2963E+09	Pa
10	Multilinear Isotropic Hardening	Tabular	
11	Scale	1	
12	Offset	0	MPa
13	Principal Stress Failure		
14	Maximum Tensile Stress	52	MPa
15	Maximum Shear Stress	35	MPa
16	Erode on Failure		

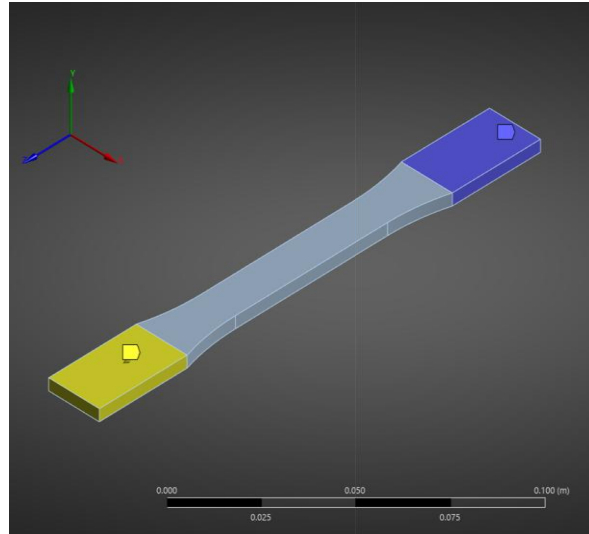
Structural Analysis: Ansys Study Setup

- Meshing appeared to be a challenge due to fine details within the models, with multiple trial and error, an extremely fine mesh with most elements within the 0-0.6 skewness value was used



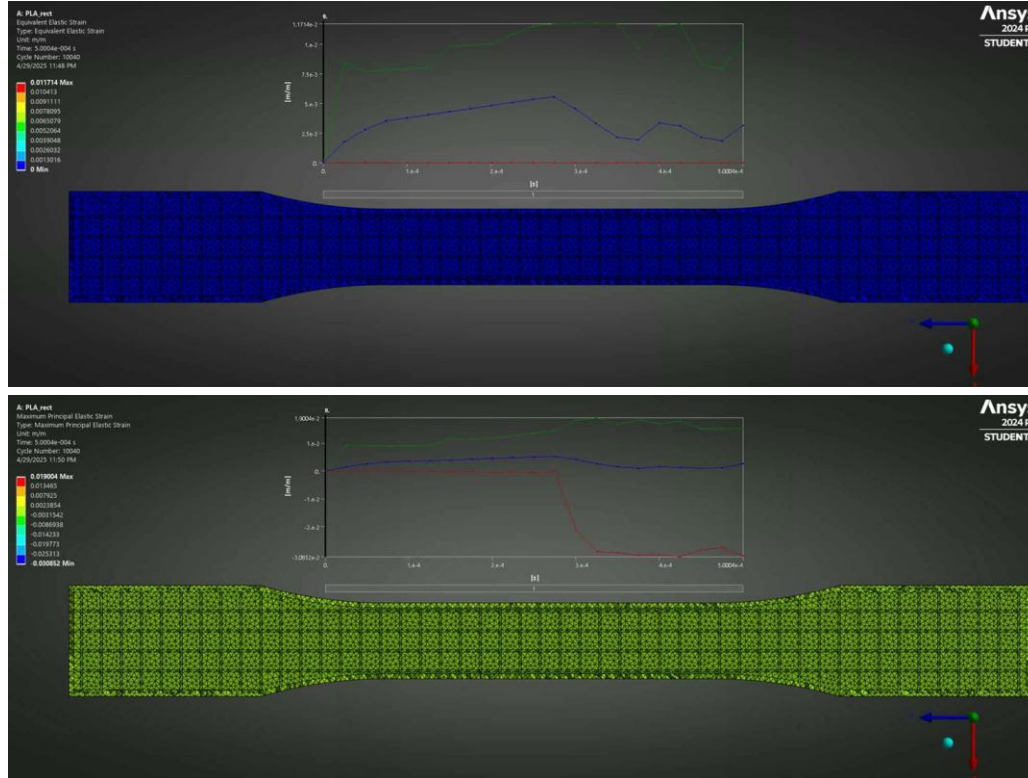
Structural Analysis: Ansys Study Setup

- Similar to actual tensile test procedure, one side was fixed and a displacement of 0.01m was applied to the the other side to obtain stress and strain results for comparison



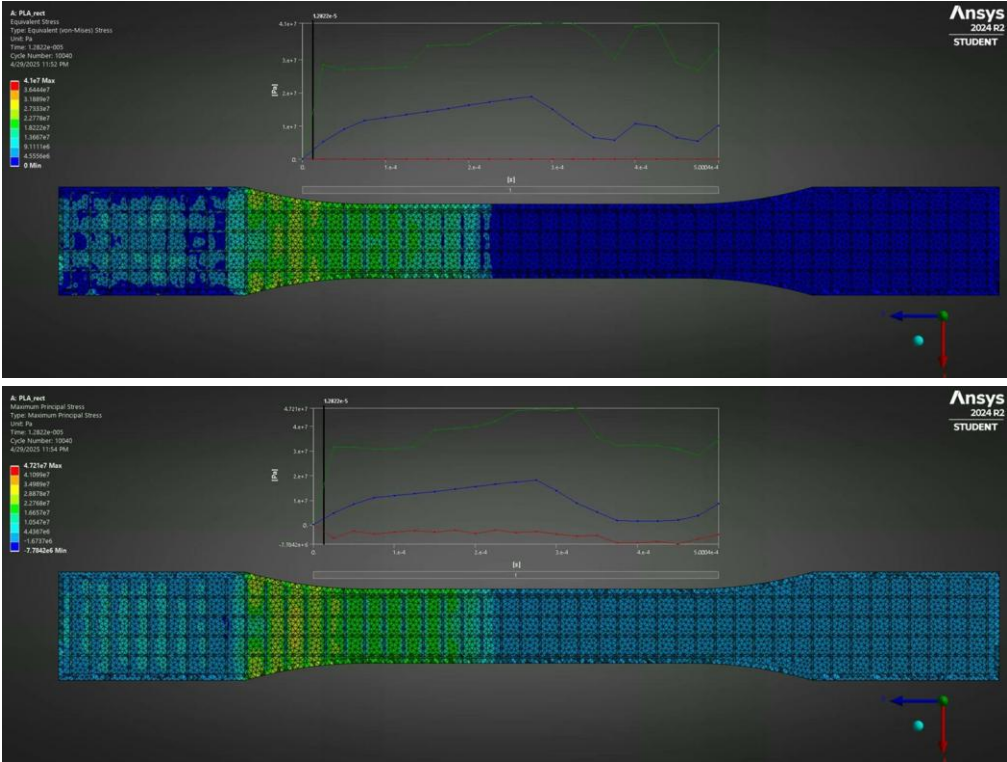
Results

Tensile Test Results: Rectilinear | Strain



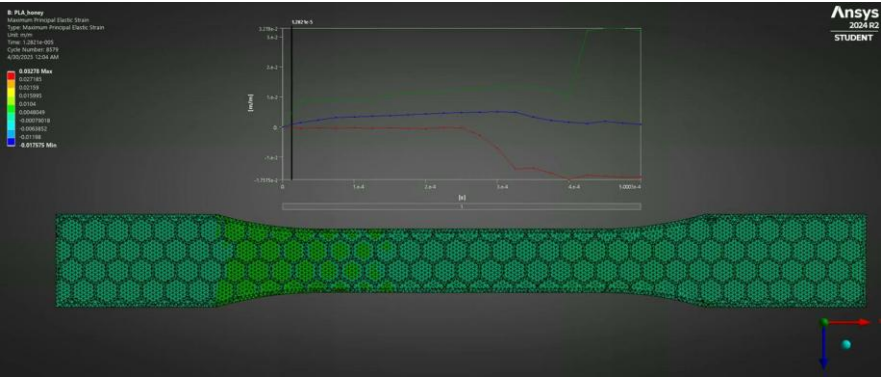
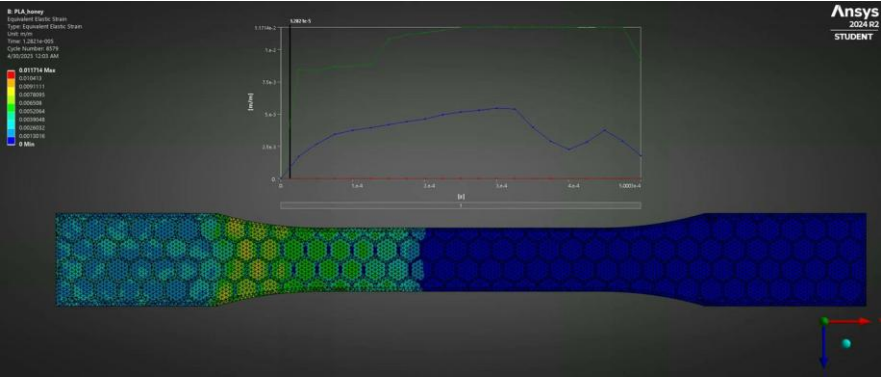
Equivalent Strain (m/m)	Maximum Principal Strain (m/m)
0.0117	0.0190

Tensile Test Results: Rectilinear | Stress



Equivalent Stress (Mpa)	Maximum Principal Stress (Mpa)
41.0	47.2

Tensile Test Results: Honeycomb | Strain



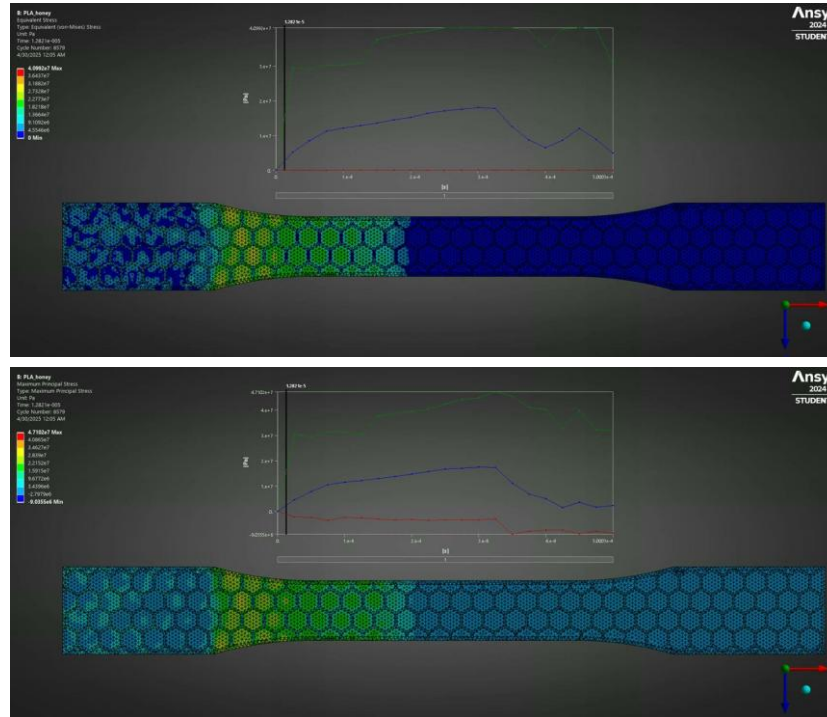
Equivalent
Strain (m/m)

Maximum
Principal Strain
(m/m)

0.0117

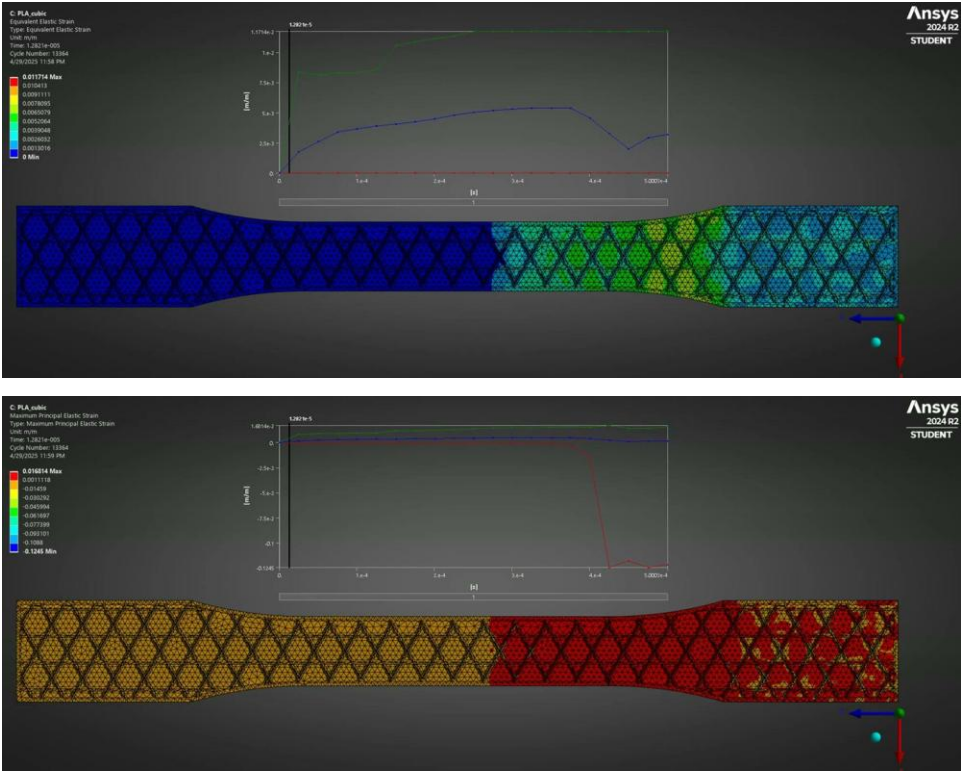
0.0328

Tensile Test Results: Honeycomb | Stress



Equivalent Stress (Mpa)	Maximum Principal Stress (Mpa)
41.0	47.1

Tensile Test Results: Cubic | Strain



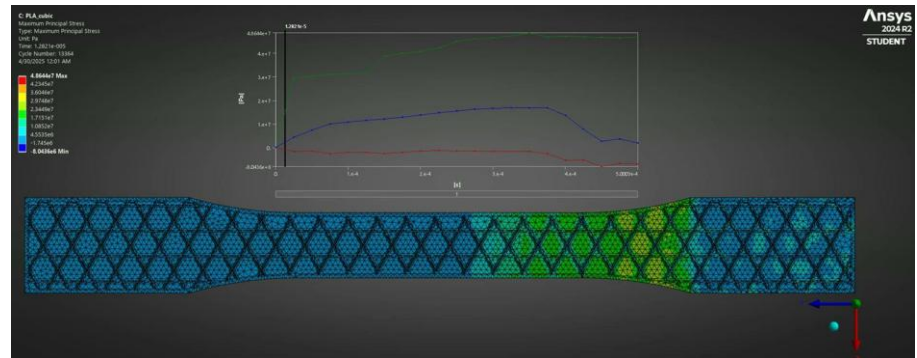
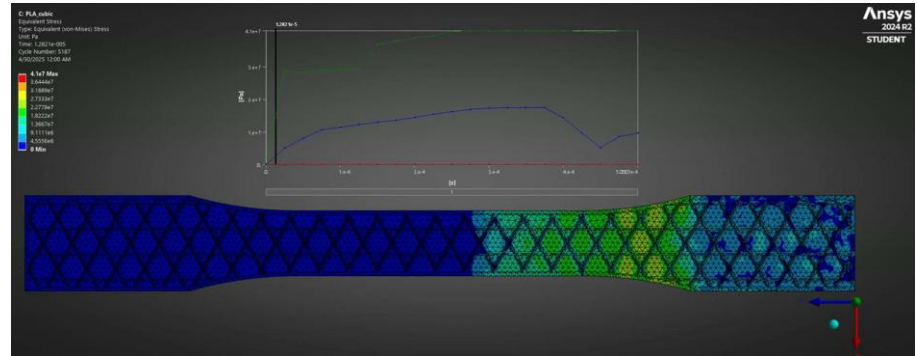
Equivalent Strain (m/m)

Maximum Principal Strain (m/m)

0.0117

0.0168

Tensile Test Results: Cubic | Stress



Equivalent Stress (Mpa)	Maximum Principal Stress (Mpa)
41.0	48.6

Tensile Test Results: Summary

- Cubic infill performs the best among the three in tensile loading, however it does not elongate as much before failure
- Honeycomb infill is best in distributing the tensile load and gives twice the strain compared to cubic infill before failure
- Rectilinear and Honeycomb infills have similar maximum principal stresses

Infill Type	Equivalent Strain (m/m)	Maximum Principal Strain (m/m)	Equivalent Stress (Mpa)	Maximum Principal Stress (Mpa)
Rectilinear	0.0117	0.0190	41.0	47.2
Honeycomb	0.0117	0.0328	41.0	47.1
Cubic	0.0117	0.0168	41.0	48.6

Tensile Test Results: Comparison with Related Literature

- A study on different infill patterns by Lin et al. entitled “Optimizing 3D printing parameters: Evaluating infill type and layer height effects on tensile fracture force” also concluded that the cubic infill, composed mostly of short and straight lines, did not elongate as much before failure compared to honeycomb, indicating higher strength but lower strain to failure.
- 2D honeycomb structure performed better than cubic in distributing the load and in terms of elongation before failure.
- Another study by Aboelella et al. entitled “Layer combination of similar infill patterns on the tensile and compression behavior of 3D printed PLA” observed that both rectilinear (grid) and honeycomb infills provide similar tensile strengths at low infill densities.

Thermal Analysis of 3D Printed Infill Patterns

How do different infill patterns affect thermal conductivity in 3D printed components?

Study Overview:

- Systematic testing of 12 different infill patterns at fixed 25% density using PLA material.
- Standardized thermal testing using hot box method.



(Source: <https://www.mdpi.com/2073-4360/15/10/2268>)

Experimental Setup - Hotbox Testing

Hotbox Apparatus:

- Constructed according to ASTM C1363-11:2011 standards
- Two five-sided climate-controlled chambers (hot & cold) with 1m³ internal volume
- 20cm thick insulated envelope($k = 0.037 \text{ W/m}\cdot\text{K}$) to reduce heat losses
- Hot chamber maintained at $32 \pm 2^\circ\text{C}$

Measurement System:

- Heat flux sensors and temperature probes on both surfaces
- Temperature measurement accuracy: $\pm 0.1\text{K}$
- Heat flux measurement accuracy: $\pm 3\%$
- Thermal resistance (R-Value) calculation: $R = \sum(T_{sij} - T_{sej}) / \sum q_j$, where
 - q_j = heat flux at time j
 - T_{sij} = inside surface temperature at time j
 - T_{sej} = outside surface temperature at time j
- Thermal conductivity = thickness of sample/ thermal resistance



Fig: Thermal test procedure using hotbox

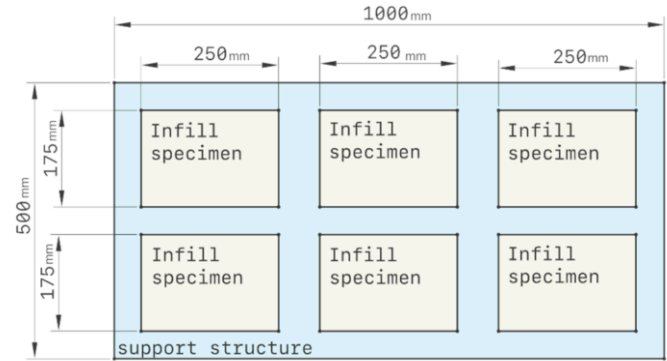
Test Specimen & Fabrication Parameters

Fabrication Parameters:

- Material: PLA filament
- Specimen size: $175 \times 250 \times 100$ mm
- Printer: Builder Extreme 1500 Pro industrial 3D printer
- Nozzle diameter: 1.2 mm
- Layer height: 0.6 mm
- Print temperature: 210°C
- Print speed: 70 mm/s
- Extrusion width: 1.26 mm

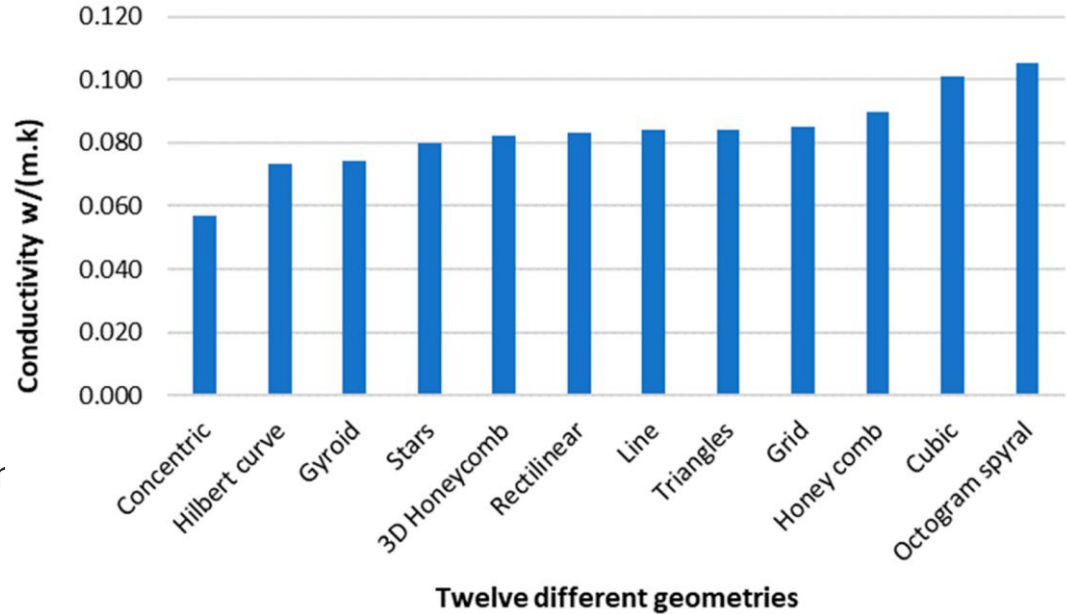
Infill Patterns Tested:

- 12 different infill patterns at 25% density



Experimental Results

- Concentric pattern has the least thermal conductivity (0.057 W/m.k)
- Octogram spiryal pattern has the highest thermal conductivity (0.105 W/m.k)
- Gyroid, Honeycomb and rectilinear pattern has mid range thermal conductivity.
- Patterns showed significant effect or thermal conductivity around 45% between pattern with lowest to highest thermal conductivity.



Conclusion

Conclusion

- Infill pattern have significant influence on the mechanical and thermal properties.

Mechanical performance:

- Cubic pattern achieved highest stress resistance(48.6 MPa).
- Honeycomb provides optimal strength-to-weight ratio.
- Rectilinear pattern showed consistent strain distribution.

Thermal performance:

- Concentric pattern: lowest conductivity (0.057 W/m·K)
- Octogram spyral has 45% higher thermal conductivity.
- Pattern density directly correlates with heat dissipation.

Conclusion

- Infill pattern and percentage: A critical balance between strength and material usage.
- Print time also varies significantly between pattern types.

Industry Applications:

- Aerospace: Lightweight honeycomb pattern suitable for structural components.
- Medical: Gyroid patterns for biocompatible implants.
- Prototyping: Rectilinear for rapid, cost-effective production.

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End of Presentation



ANY QUESTIONS?

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Dilli Babu
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